```python
import numpy as np
from scipy.integrate import solve_ivp

a = 0.1
b = 0.001
x0 = 2000

def analytical_solution_bacteria(t, x0, a, b):

    num = x0 * a * np.exp(a * t)
    d_num = (a - b * x0) + b * x0 * np.exp(a * t)
    return num / d_num

def bac_growth_ode(t, x):
    return a * x - b * x**2

x0_bacteria = 2000
t0_bacteria = 0
total_time_bacteria = 4

t_span = (t0_bacteria, total_time_bacteria)
t_eval = [total_time_bacteria]

solution_ivp = solve_ivp(bac_growth_ode, t_span, [x0_bacteria],
t_eval=t_eval, method='RK45')
numerical_bacteria_at_4h = solution_ivp.y[0][-1]
analytical_bacteria_at_4h =
analytical_solution_bacteria(total_time_bacteria, x0, a, b)

if analytical_bacteria_at_4h != 0:
    percentage_error = abs((numerical_bacteria_at_4h -
analytical_bacteria_at_4h) / analytical_bacteria_at_4h) * 100
else:
    percentage_error = float('inf')

print(f"\nAnalytical solution for bacteria after {total_time_bacteria}
hours: {analytical_bacteria_at_4h:.2f}")
print(f"Numerical solution (SciPy solve_ivp) for bacteria after
{total_time_bacteria} hours: {numerical_bacteria_at_4h:.2f}")
print(f"Percentage Error: {percentage_error:.4f}%")
```

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Analytical solution for bacteria after 4 hours: 275.33
Numerical solution (SciPy solve_ivp) for bacteria after 4 hours:
275.82
Percentage Error: 0.1784%
```